

CSE552: Business Analytics

Unit 2: Inferential Statistics and Hypothesis Testing

Lecture Notes

Session 2026-27

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2.1 Normal Distribution

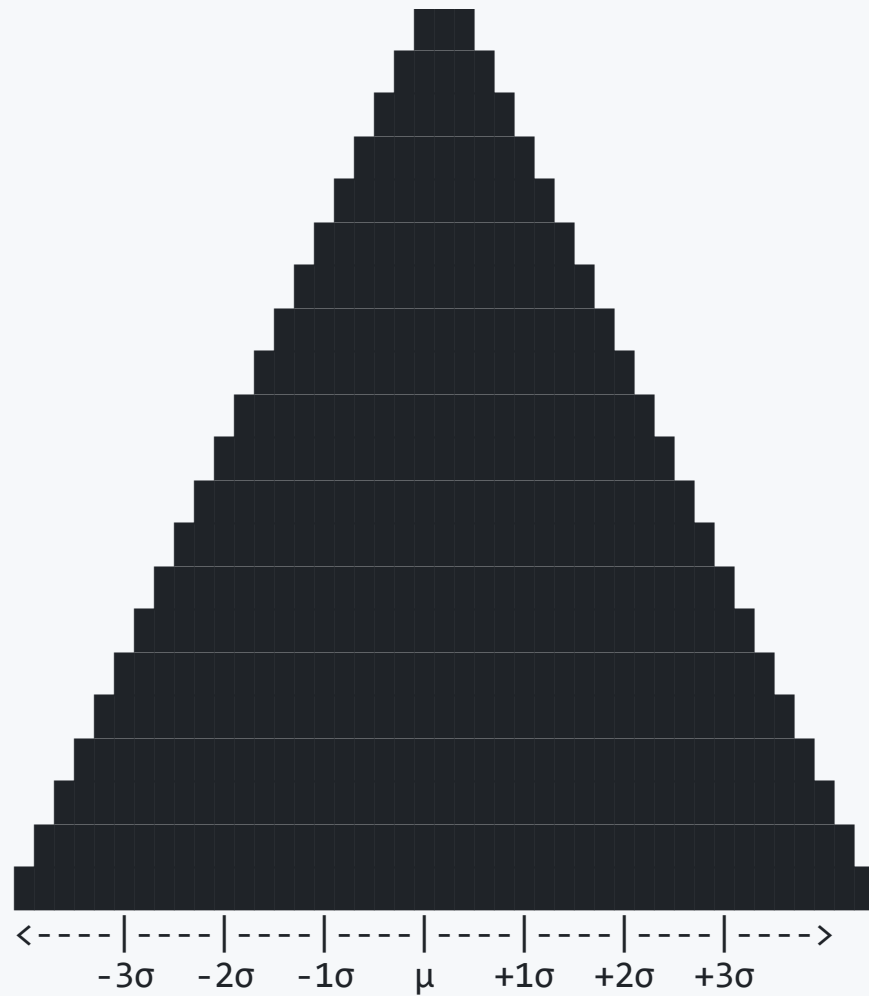
Definition

Normal Distribution (also called Gaussian distribution) is a continuous probability distribution that is symmetric about its mean, showing that data near the mean are more frequent in occurrence than data far from the mean. It is the most important probability distribution in statistics because many natural phenomena follow this pattern.

Why is Normal Distribution Important?

- **Natural Phenomena:** Heights, weights, IQ scores, measurement errors
- **Statistical Foundation:** Basis for many statistical tests
- **Central Limit Theorem:** Sample means tend to follow normal distribution
- **Predictability:** Allows us to make probability statements about data

The Bell Curve

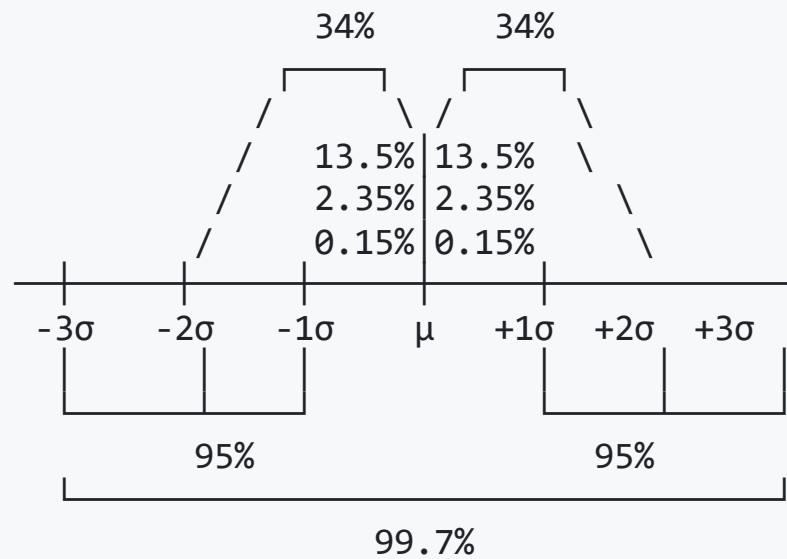


Properties of Normal Distribution

Property	Description
Symmetry	Perfectly symmetric about the mean (μ)
Mean = Median = Mode	All three measures coincide at the center
Asymptotic	Tails approach but never touch the x-axis
Area = 1	Total area under the curve equals 1 (100%)
Defined by μ and σ	Mean (μ) and Standard Deviation (σ) completely define the curve

The Empirical Rule (68-95-99.7 Rule)

- 68% of data falls within $\mu \pm 1\sigma$
- 95% of data falls within $\mu \pm 2\sigma$
- 99.7% of data falls within $\mu \pm 3\sigma$



Probability Density Function (PDF)

Formula:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Where:

- μ (mu) = mean (location parameter)
- σ (sigma) = standard deviation (scale parameter)
- $e \approx 2.71828$ (Euler's number)
- $\pi \approx 3.14159$

Standard Normal Distribution

Standard Normal Distribution is a special case of normal distribution where:

- Mean (μ) = 0
- Standard Deviation (σ) = 1

Any normal distribution can be converted to standard normal using the **Z-score transformation**.

A z-score, also called a standard score, indicates the position of a data point relative to the mean of a distribution in terms of standard deviations.

- A z-score of 0 means the value is equal to the mean, a positive z-score indicates the value is above the mean, and a negative z-score indicates it is below the mean.
- Z-scores are useful for comparing values from different distributions or identifying outliers

Solved Problem: Normal Distribution

Problem: The heights of students in a college are normally distributed with a mean of 170 cm and a standard deviation of 10 cm. Find the probability that a randomly selected student:

- (a) Is taller than 185 cm?
- (b) Is between 160 cm and 180 cm?

Given:

(Z) -table, $(P(Z \leq 1.5) = 0.9332)$.

1. Identify the given parameters

The heights of the students (X) follow a normal distribution described by:

- Mean (μ) = (170 cm)
- Standard deviation (σ) = (10 cm)

To solve these problems, any data point (X) must be converted into a standard normal score (Z -score) using the transformation formula:

$$(Z = \frac{X - \mu}{\sigma})$$

- Steps to normally distributed dataset by converting it to the standard normal form using the z-score formula.
- This standardization process, also known as z-score normalization, ensures that all features are on the same scale, which is crucial for many machine learning algorithms that rely on distance-based calculations.
- It helps in identifying outliers, as extreme z-scores often indicate anomalies in the data.
- Additionally, it standardizes the data for comparisons, preventing features with larger ranges from disproportionately influencing results.
- This makes it easier to interpret and analyze the data effectively.

2. Calculate for part (a)

Find the probability that a randomly selected student: (a) Is taller than 185 cm?

We want to find $(P(X > 185))$.

First, calculate the corresponding (Z)-score for $(X = 185 \text{ cm})$:

$$(Z = \frac{185-170}{10} = \frac{15}{10} = 1.5)$$

Now, find the area to the right of $(Z = 1.5)$ using a standard normal distribution table:

$$P(X > 185) = P(Z > 1.5) = 1 - P(Z \leq 1.5))$$

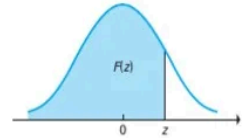
To be Given: From the standard (Z) -table, $(P(Z \leq 1.5) = 0.9332)$.

Therefore:

$$(P(Z > 1.5) = 1 - 0.9332 = 0.0668)$$

APPENDIX TABLES

Table 1 Cumulative Distribution Function, $F(z)$, of the Standard Normal Distribution Table



z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997

Dr. William L. Carlson, prepared using Minitab 16.

Table A.3 Normal Probability Table

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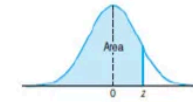


Table A.3 Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

3. Calculate for part (b)

Find the probability that a randomly selected student: (b) Is between 160 cm and 180 cm?

We want to find $(P(160 < X < 180))$.

First, calculate the (Z) -scores for both bounds:

$$\text{For } (X_1 = 160 \text{ cm}): (Z_1 = \frac{160-170}{10} = \frac{-10}{10} = -1.0) \text{ For } (X_2 = 180 \text{ cm}) : (Z_2 = \frac{180-170}{10} = \frac{10}{10} = 1.0)$$

Now, calculate the probability between these values:

$$(P(160 < X < 180) = \underline{P(-1.0 < Z < 1.0)} = P(Z \leq 1.0) - P(Z \leq -1.0))$$

$$[\text{Given}] \text{ From the standard } (Z) - \text{table} : (P(Z \leq 1.0) = \underline{0.8413})(P(Z \leq -1.0) = \underline{0.1587})$$

$$\text{Subtract the two values: } (P(-1.0 < Z < 1.0) = 0.8413 - 0.1587 = 0.6826)$$

(Note: According to the empirical rule, exactly 1 standard deviation contains approximately (68.27%) of the data).

Convert the probability to a percentile by multiplying by 100 if needed.

Solution (Part a) Simplified

Given: $\mu = 170$ cm, $\sigma = 10$ cm

Step 1: Calculate Z-score for $x = 185$

$$Z = \frac{X - \mu}{\sigma} = \frac{185 - 170}{10} = \frac{15}{10} = 1.5$$

Step 2: Find $P(Z > 1.5)$

From Z-table: $P(Z < 1.5) = 0.9332$

$$P(X > 185) = P(Z > 1.5) = 1 - 0.9332 = 0.0668$$

Answer: 6.68% probability (about 1 in 15 students)

Solution (Part b) Simplified

Step 1: Calculate Z-scores

$$\text{For } x = 160: Z = \frac{160-170}{10} = -1.0$$

$$\text{For } x = 180: Z = \frac{180-170}{10} = +1.0$$

Step 2: Find $P(-1 < Z < 1)$

From Z-table: $P(Z < 1) = 0.8413$, $P(Z < -1) = 0.1587$

$$P(160 < X < 180) = P(-1 < Z < 1) = 0.8413 - 0.1587 = 0.6826$$

Answer: 68.26% probability (matches the Empirical Rule!)

2.2 Central Limit Theorem (CLT)

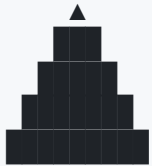
Definition

Central Limit Theorem (CLT) states that given a sufficiently large sample size (typically $n \geq 30$), the sampling distribution of the mean for a variable will approximate a normal distribution regardless of the variable's distribution in the population.

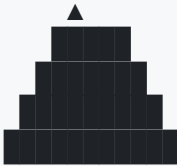
Why CLT is Revolutionary

Population Distribution (Any Shape!)

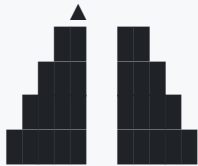
Skewed:



Uniform:

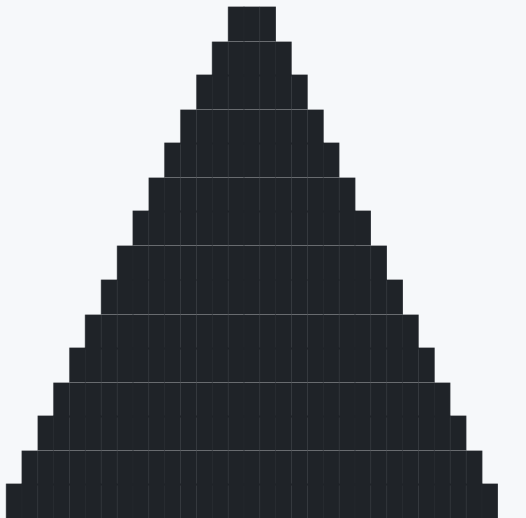


Bimodal:



After taking samples ($n \geq 30$) and computing means:

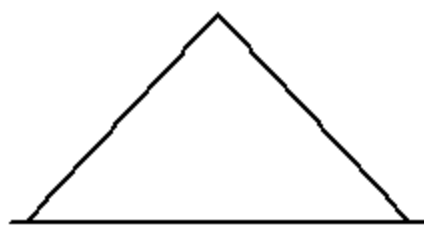
Sampling Distribution of Sample Means:



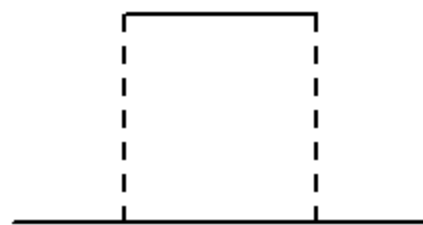
NORMAL DISTRIBUTION!



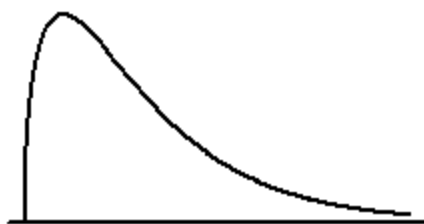
(a) Bell shaped



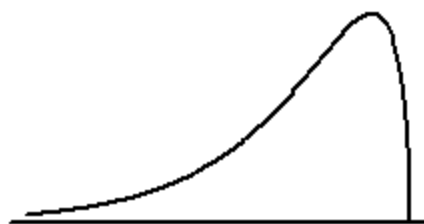
(b) Triangular



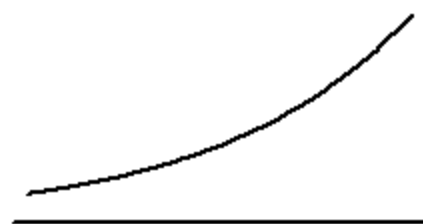
(c) Uniform (rectangular)



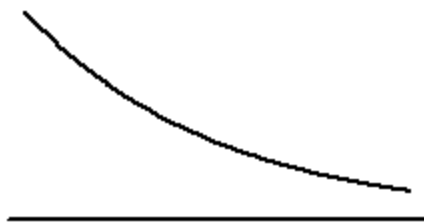
(d) Right skewed



(e) Left skewed



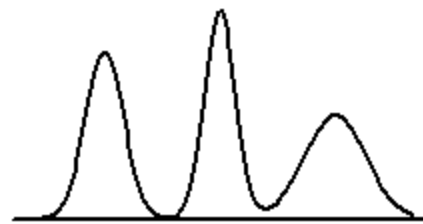
(f) J Shaded



(g) Reverse J Shaded



(h) Bimodal



(i) Multi-modal

Key Points of CLT

1. **Original population can be ANY distribution** (skewed, uniform, bimodal, etc.)
2. **Sample size matters:** $n \geq 30$ is the rule of thumb
3. **Larger sample size** = better approximation to normal
4. **Applies to sample MEANS**, not individual observations
5. **Foundation of inferential statistics** : allows you to make predictions or generalizations about a broader population based on a smaller.

CLT Formulas

Sampling Distribution of the Mean:

- Mean of sample means: $\mu_{\bar{x}} = \mu$ (same as population mean)
- Standard Error (SE): $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$

Z-score for Sample Mean:

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

Solved Problem: CLT

Problem: A population has mean $\mu = 50$ and standard deviation $\sigma = 12$. A random sample of 36 observations is taken. What is the probability that the sample mean is greater than 52?

Solution

Given: $\mu = 50$, $\sigma = 12$, $n = 36$, $\bar{x} = 52$

Step 1: Calculate Standard Error

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{12}{\sqrt{36}} = \frac{12}{6} = 2$$

Step 2: Calculate Z-score for sample mean

$$Z = \frac{\bar{X} - \mu}{SE} = \frac{52 - 50}{2} = \frac{2}{2} = 1.0$$

Solution (Continued)

Step 3: Find probability

$$\begin{aligned} P(Z > 1.0) \\ &= 1 - P(Z < 1.0) \\ &= 1 - 0.8413 \\ &= \mathbf{0.1587} \end{aligned}$$

Answer: There is a **15.87%** probability that the sample mean exceeds 52.

Interpretation: Even though 52 is close to the population mean (50), only about 16% of samples of size 36 would have a mean this high or higher. This is because the standard error (2) is smaller than the population standard deviation (12).

2.3 Population vs Samples

Definitions

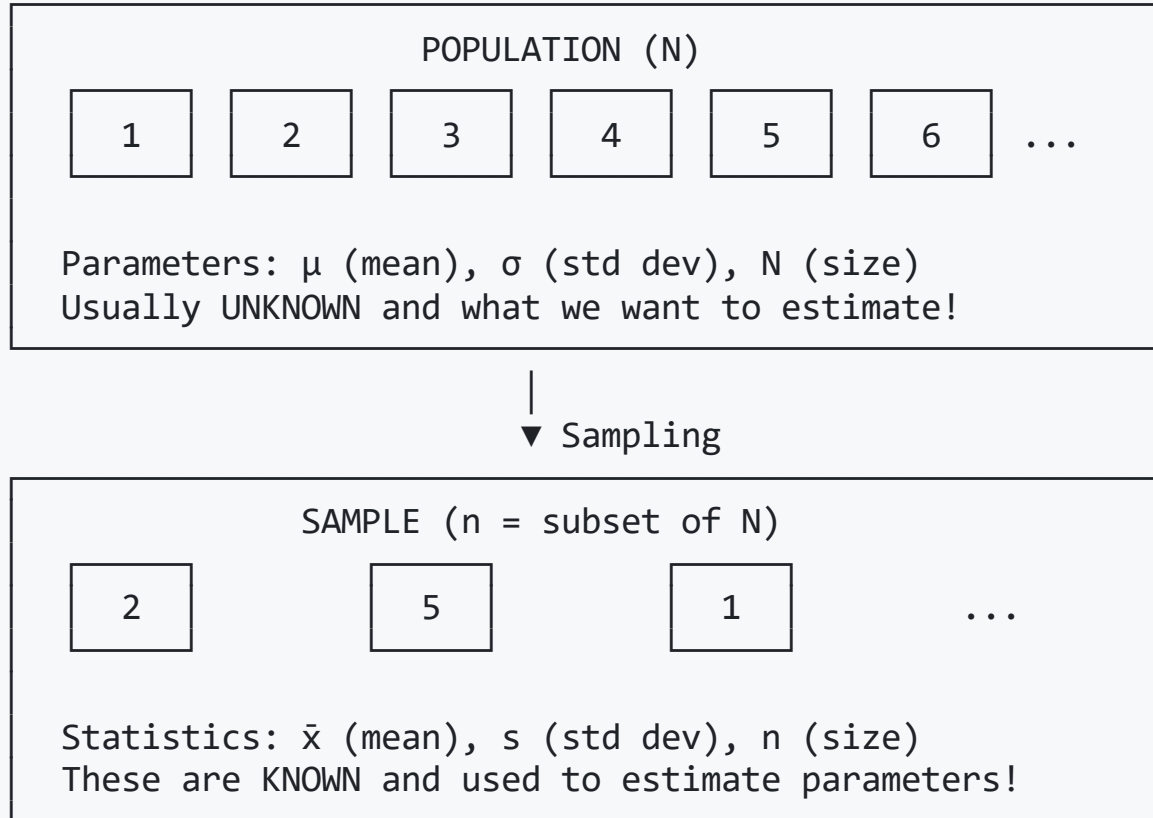
Population: The entire group of individuals or items that we want to study and draw conclusions about.

- **Population Size:** Denoted by N
- **Population Mean:** Denoted by μ (mu)
- **Population Standard Deviation:** Denoted by σ (sigma)
- **Population Parameter:** A numerical value that describes a characteristic of the population

Sample: A subset of the population selected for study to make inferences about the population.

- **Sample Size:** Denoted by n
- **Sample Mean:** Denoted by $\bar{x}$ (x-bar)
- **Sample Standard Deviation:** Denoted by s
- **Sample Statistic:** A numerical value calculated from sample data

Population vs Sample: Visual Comparison



Why Use Samples?

Reason	Explanation
Cost	Studying entire population is expensive
Time	Faster to analyze a sample
Destructive Testing	Testing destroys the item (e.g., light bulb lifespan)
Accessibility	Population may be impossible to reach entirely
Accuracy	Better to have quality data from a sample than poor data from all

2.4 Sampling Distributions

Definition

Sampling Distribution is the probability distribution of a statistic (like mean, proportion, variance) obtained from a large number of samples drawn from a specific population.

It describes how the statistic varies from sample to sample.

- A sampling distribution is the probability distribution of a statistic (like the mean or proportion) obtained from multiple random samples of the same population.
- It is a core concept in inferential statistics, enabling us to estimate population parameters and perform hypothesis testing.

Key terms

- **Statistic:** Value from a sample (e.g., sample mean $\bar{x}$).
- **Parameter:** Value from the population (e.g., μ).
- **Standard Error (SE):** Standard deviation of the sampling distribution.
- **Central Limit Theorem (CLT):** For large sample sizes, the sampling distribution of the mean approaches normality, regardless of the population's distribution.

Main Types of Sampling Distributions

- Sampling Distribution of the Mean Mean of sampling distribution: $\mu_{\bar{x}} = \mu$
- Standard deviation: $\sigma_{\bar{x}} = \sigma/\sqrt{n}$
- Unbiased: Sample mean does not systematically over/underestimate μ .
- Sampling Distribution of the Proportion Formula: $\hat{p} = x/n$ $\hat{p}$ is the proportion of successes in a sample. Useful for categorical data analysis.
- T-Distribution Used when population standard deviation is unknown or sample size is small. Formula: $t = (\bar{x} - \mu)/(s/\sqrt{n})$ Critical for confidence intervals and hypothesis testing with small samples.

Types of Sampling Distributions

SAMPLING DISTRIBUTIONS
1. SAMPLING DISTRIBUTION OF MEAN → Distribution of sample means ($\bar{x}$) → Used for: Estimating population mean 2. SAMPLING DISTRIBUTION OF PROPORTION → Distribution of sample proportions ($\hat{p}$) → Used for: Estimating population proportion 3. SAMPLING DISTRIBUTION OF VARIANCE → Distribution of sample variances (s^2) → Used for: Testing hypotheses about variance

Sampling Distribution of the Mean

Properties:

- Mean of sampling distribution: $\mu_{\bar{x}} = \mu$
- Standard Error: $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$
- Shape: Normal (by CLT, when $n \geq 30$)

Key Insight: As sample size (n) increases, standard error decreases, meaning sample means cluster more tightly around the population mean.

Visual: Effect of Sample Size

Population Distribution ($\sigma = 10$)

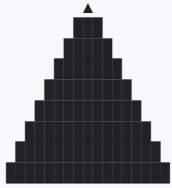


Sampling Distribution ($n = 4$, $SE = 5$)



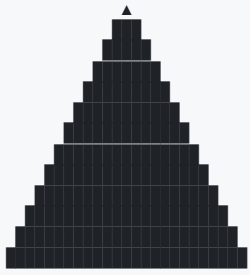
Wider spread

Sampling Distribution ($n = 25$, $SE = 2$)



Narrower spread
More precise!

Sampling Distribution ($n = 100$, $SE = 1$)



Very precise!

2.5 Z-Score

Definition

Z-Score (or Standard Score) indicates how many standard deviations a data point is from the mean of the distribution.

It standardizes values from different distributions to a common scale.

Z-Score Formula

For Individual Data Point:

$$Z = \frac{X - \mu}{\sigma}$$

For Sample Mean:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

Interpretation:

- $Z > 0$: Above average
- $Z < 0$: Below average
- $Z = 0$: Exactly average
- $|Z| > 2$: Unusual (only 5% of data)
- $|Z| > 3$: Very unusual (only 0.3% of data)

Z-Score Interpretation Scale

Z-Score Scale:



Common Range: -2 to +2 (95% of data)

Solved Problem: Z-Score

Problem: In an exam, the mean score is 75 with a standard deviation of 8.

- (a) What is the Z-score of a student who scored 91?
- (b) What score corresponds to a Z-score of -1.5?
- (c) What percentage of students scored between 67 and 83?

Solution (Part a)

$$Z = \frac{X - \mu}{\sigma} = \frac{91 - 75}{8} = \frac{16}{8} = 2.0$$

Interpretation: The student scored **2 standard deviations above the mean**. This is an excellent score (top ~2.3% of the class).

Solution (Part b)

Rearranging the Z-score formula:

$$X = \mu + Z \times \sigma = 75 + (-1.5) \times 8 = 75 - 12 = 63$$

Answer: A Z-score of -1.5 corresponds to a score of **63**.

Solution (Part c)

$$\text{For } X = 67: Z = \frac{67-75}{8} = -1.0$$

$$\text{For } X = 83: Z = \frac{83-75}{8} = +1.0$$

From Empirical Rule: $P(-1 < Z < 1) = 68\%$

Answer: Approximately **68%** of students scored between 67 and 83.

This makes sense because $67 = \mu - \sigma$ and $83 = \mu + \sigma$, and by the Empirical Rule, 68% of data falls within 1 standard deviation of the mean.

2.6 P-Value

Definition

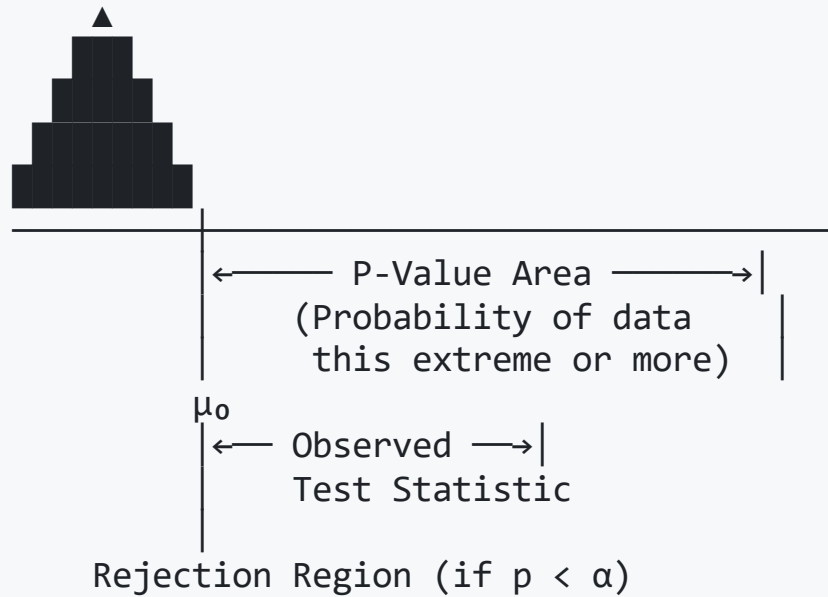
P-Value (Probability Value) is the probability of obtaining test results at least as extreme as the observed results, assuming that the null hypothesis is true.

In simpler terms: "What is the probability of seeing this data (or more extreme) if the null hypothesis were actually true?"

Understanding P-Value

P-Value Visual Representation:

Null Hypothesis Distribution



P-Value Decision Rule

Decision Rule:

- If **p-value $\leq \alpha$** (significance level, typically 0.05): **Reject H_0**
 - Result is "statistically significant"
 - Evidence supports the alternative hypothesis
- If **p-value $> \alpha$** : **Fail to Reject H_0**
 - Result is "not statistically significant"
 - Insufficient evidence to reject the null

Common Significance Levels:

- $\alpha = 0.10$ (10%): Weak evidence
- $\alpha = 0.05$ (5%): Moderate evidence (most common)
- $\alpha = 0.01$ (1%): Strong evidence

P-Value Interpretation Guide

P-Value Range	Evidence Against H_0	Interpretation
$p > 0.10$	No evidence	Not significant
$0.05 < p \leq 0.10$	Weak evidence	Marginally significant
$0.01 < p \leq 0.05$	Moderate evidence	Significant
$0.001 < p \leq 0.01$	Strong evidence	Highly significant
$p \leq 0.001$	Very strong evidence	Extremely significant

Important Note About P-Values

What P-Value is NOT:

- ✗ NOT the probability that the null hypothesis is true
- ✗ NOT the probability that the alternative hypothesis is false
- ✗ NOT the probability that the results occurred by chance alone
- ✗ NOT a measure of practical significance

What P-Value IS:

- ✓ The probability of observing data this extreme (or more) IF H_0 were true
- ✓ A measure of compatibility between data and the null hypothesis
- ✓ A tool for making decisions about hypotheses

2.7 Rejection Region

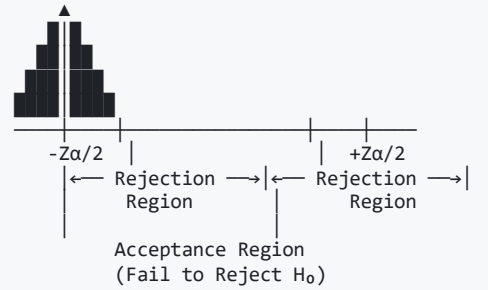
Definition

Rejection Region (or Critical Region) is the set of values for the test statistic that leads to rejection of the null hypothesis.

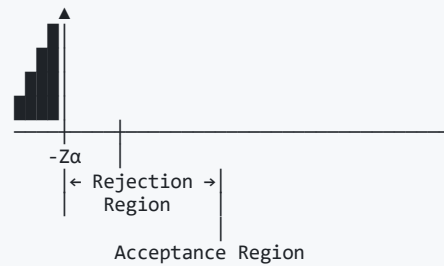
If the test statistic falls in this region, we conclude that the observed data is too unlikely under the null hypothesis.

Types of Tests and Rejection Regions

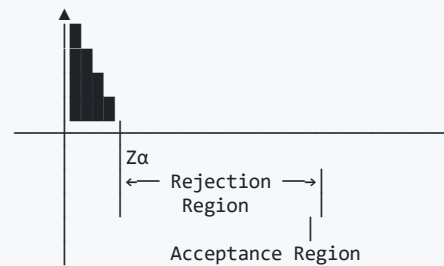
TWO-TAILED TEST ($H_1: \mu \neq \mu_0$)



ONE-TAILED TEST (Left: $H_1: \mu < \mu_0$)



ONE-TAILED TEST (Right: $H_1: \mu > \mu_0$)



Critical Values Table

Common Critical Values (Z-test):

Test Type	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.01$
Two-tailed	± 1.645	± 1.96	± 2.576
Left-tailed	-1.282	-1.645	-2.326
Right-tailed	+1.282	+1.645	+2.326

2.8 Hypothesis Testing

Definition

Hypothesis Testing is a statistical method used to make decisions about a population parameter based on sample data. It involves:

1. Making an assumption (hypothesis) about a population parameter
2. Collecting sample data
3. Determining whether the sample data supports or contradicts the assumption

The Two Hypotheses

Null Hypothesis (H_0):

- The statement being tested (assumed true)
- Usually contains "=", " $\leq$ ", or " $\geq$ "
- Represents "no effect", "no difference", "status quo"
- **Example:** $H_0: \mu = 100$ (The average IQ is 100)

Alternative Hypothesis (H_1 or H_a):

- The statement we want to find evidence FOR
- Usually contains " $\neq$ ", "<", or ">"
- Represents "there is an effect", "there is a difference"
- **Example:** $H_1: \mu \neq 100$ (The average IQ is not 100)

Types of Hypothesis Tests

TYPES OF HYPOTHESIS TESTS

TWO-TAILED TEST:

$H_0: \mu = \mu_0$ vs $H_1: \mu \neq \mu_0$

"Is there ANY difference?"

Rejection regions: BOTH tails

LEFT-TAILED TEST:

$H_0: \mu \geq \mu_0$ vs $H_1: \mu < \mu_0$

"Is it significantly LESS?"

Rejection region: LEFT tail only

RIGHT-TAILED TEST:

$H_0: \mu \leq \mu_0$ vs $H_1: \mu > \mu_0$

"Is it significantly MORE?"

Rejection region: RIGHT tail only

Steps in Hypothesis Testing

Step 1: State the null hypothesis (H_0) and alternative hypothesis (H_1)

Step 2: Choose the significance level (α), typically 0.05

Step 3: Select the appropriate test statistic (Z-test, t-test, etc.)

Step 4: Calculate the test statistic from sample data

Step 5: Determine the critical value or p-value

Step 6: Make a decision (Reject H_0 or Fail to Reject H_0)

Step 7: State the conclusion in the context of the problem

Errors in Hypothesis Testing

DECISION TABLE		
DECISION	ACTUAL TRUTH	
	H_0 is TRUE	H_0 is FALSE
Reject H_0	Type I Error (False Positive) Probability = α	Correct Decision (True Positive) Power = $1-\beta$
Fail to Reject H_0	Correct Decision (True Negative) Probability= $1-\alpha$	Type II Error (False Negative) Probability = β

Type I and Type II Errors Explained

Type I Error (α):

- Rejecting H_0 when it is actually TRUE
- "False Positive" - Finding an effect that doesn't exist
- **Example:** Concluding a drug works when it actually doesn't
- Controlled by significance level α

Type II Error (β):

- Failing to reject H_0 when it is actually FALSE
- "False Negative" - Missing an effect that does exist
- **Example:** Concluding a drug doesn't work when it actually does
- Power of test = $1 - \beta$ (probability of correctly detecting an effect)

Trade-off: Decreasing α increases β , and vice versa.

Solved Problem: Hypothesis Testing (Z-Test)

Problem: A company claims that their light bulbs last an average of 1000 hours. A consumer group believes the average is less. They test a sample of 50 bulbs and find a mean of 980 hours with a standard deviation of 60 hours. Test at $\alpha = 0.05$ significance level.

Solution

Step 1: State hypotheses

- $H_0: \mu \geq 1000$ hours (Bulbs last at least 1000 hours)
- $H_1: \mu < 1000$ hours (Bulbs last less than 1000 hours)

Left-tailed test

Step 2: Significance level $\alpha = 0.05$

Step 3: Use Z-test ($n = 50 > 30$, population σ unknown but large sample)

Solution (Continued)

Step 4: Calculate test statistic

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{980 - 1000}{60/\sqrt{50}} = \frac{-20}{60/7.071} = \frac{-20}{8.485} = -2.357$$

Step 5: Critical value for left-tailed test at $\alpha = 0.05$: $Z = -1.645$

Step 6: Decision

Since $-2.357 < -1.645$ (test statistic falls in rejection region)

Decision: Reject H_0

Step 7: Conclusion

At the 0.05 significance level, there is sufficient evidence to conclude that the average lifespan of the light bulbs is **less than 1000 hours**. The company's claim is not supported by the data.

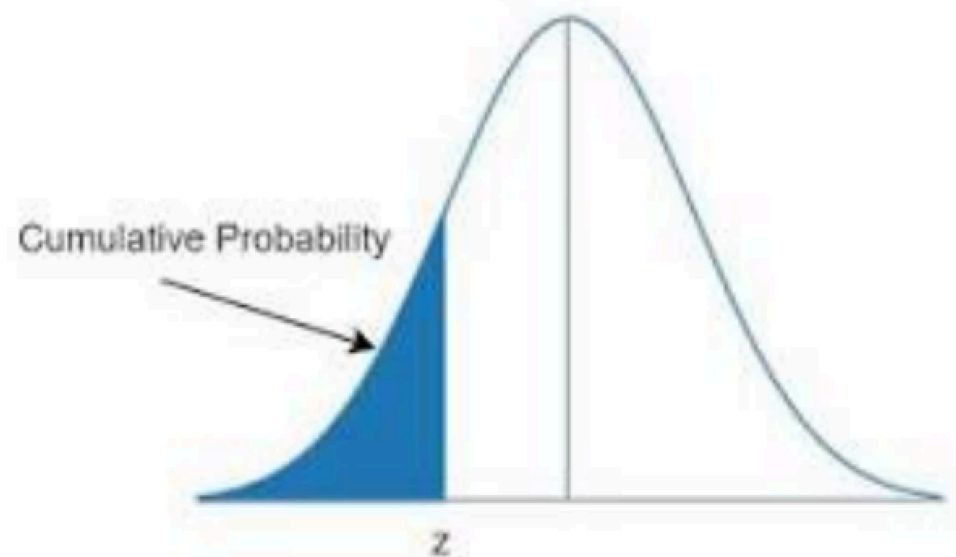
Alternative: Using P-Value Approach

Step 5 (P-value): $P(Z < -2.357) = 0.0092$ (from Z-table)

Step 6: Since p-value (0.0092) $< \alpha$ (0.05)

Decision: Reject H_0

The p-value of 0.0092 means there is only a 0.92% chance of observing a sample mean of 980 hours or less if the true mean were actually 1000 hours. This is very unlikely, so we reject the null hypothesis.



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143

Summary: Key Formulas

Concept	Formula
Z-Score (Individual)	$Z = \frac{X - \mu}{\sigma}$
Z-Score (Sample Mean)	$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$
Standard Error	$SE = \frac{\sigma}{\sqrt{n}}$
Confidence Interval	$\bar{X} \pm Z_{\alpha/2} \times SE$
Test Statistic (Z)	$Z = \frac{\text{Statistic} - \text{Parameter}}{\text{Standard Error}}$

Unit 2 Summary

Key Takeaways:

1. **Normal Distribution:** Bell-shaped, symmetric, defined by μ and σ
2. **CLT:** Sample means become normally distributed as n increases
3. **Sampling:** We use samples to infer population parameters
4. **Z-Score:** Standardizes values for comparison
5. **P-Value:** Probability of extreme data if H_0 is true
6. **Hypothesis Testing:** Structured method for making statistical decisions

Remember: Statistics is about making decisions under uncertainty. We never "prove" anything with 100% certainty - we measure the strength of evidence!

Practice Questions

1. Explain why the Central Limit Theorem is important even if the population distribution is not normal.
2. A sample of 64 observations has mean 25 and standard deviation 8. Calculate the standard error and find the probability that the sample mean exceeds 26.
3. Define Type I and Type II errors with real-world examples.
4. In a hypothesis test, what does a p-value of 0.03 mean in practical terms?
5. Why do we say "fail to reject H_0 " instead of "accept H_0 "?

(Right-tailed)

Z-Test Hypothesis Testing

Q) A machine fills a cereal boxes; the label claim $\mu = 368g$. Population std. dev $\sigma = 15g$. The quality inspector take sample $n = 25$ boxes and found $\bar{x} = 372.5g$

(A) Test whether the machine is overfilling at $\alpha = 0.05$. Assuming the cereal box weight is normally distributed

Sol $\mu = 368g$; $\sigma = 15g$; $n = 25$; $\bar{x} = 372.5g$; $\alpha = 0.05$

Step 1: Purpose Hypothesis

$$H_0: \mu = 368g (\mu = \mu_0)$$

$$H_a: \mu > 368g (\mu > \mu_0)$$



Step 2: Decide the test (left / Right / 2-tailed)

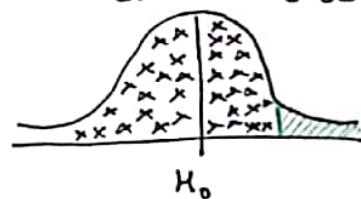
$\mu > \mu_0 \Rightarrow$ Right-tailed test

Step 3: Calculate Z & Z_c

$$(i) Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{372.5 - 368}{15 / \sqrt{25}}$$

$$= \frac{4.5}{15/5} = \frac{4.5}{3} = 1.5 \Rightarrow Z = 1.5$$

$$(ii) Z_c \Rightarrow Z_{0.05}$$

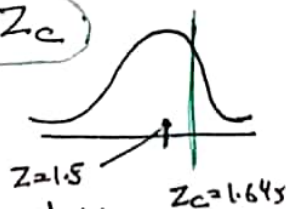


Area under the curve
($1 - 0.05 = 0.95$)

$$Z_c @ 0.95 = 1.645$$

Step 4: Compare Z & Z_c

$$Z = 1.5 ; Z_c = 1.645$$



As $Z < Z_c$: Fail to Reject H_0

Step 5: Compare probability (P-value) with significant level α

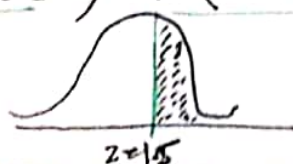
$$\alpha = 0.05 ; P(Z \geq 1.5) = P(Z > 1.5) = 0.9332$$

$$P(Z \geq 1.5) = 1 - 0.9332 = 0.0668$$

As $P(Z \geq 1.5) > \alpha \Rightarrow$ Fail to Reject H_0

Step 6 Conclusion

There is insufficient evidence @ 5% significant level to α



(Left tailed)

Z-Test Hypothesis Testing

(Large Sample)

- (B) $\bar{x} = 365$; Test whether the machine is underfilling at $\alpha = 0.05$, (Assuming weight is normally distributed)

$$\mu = 368g; \sigma = 15g; n = 25; \bar{x} = 365.5; \alpha = 0.05$$

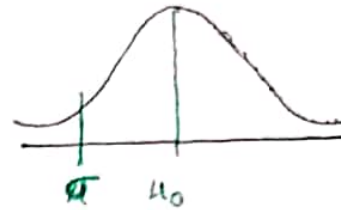
Solution:

Step 1: Draft Hypothesis

$$H_0: \mu = 368g \quad (\mu = \mu_0)$$

$$H_A: \mu < 368g \quad (\mu < \mu_0)$$

(Left-tailed Test)



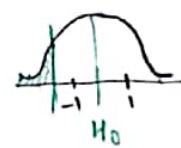
Step 2: Decide the test: Left-tailed Test

Step 3: Calculate Z & Z_c

$$(i) \quad Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{365.5 - 368}{15 / \sqrt{25}} \\ = \frac{-2}{15/5} = \frac{-2}{3} = -0.8333$$

~~$Z = -0.8333$~~ $Z = -0.8333$

$$(ii) \quad Z_c = Z_{0.05}$$



Find area under curve
@ $1 - 0.05 = 0.95$

$$Z_c = -1.645$$

Step 4: Compare Z & Z_c

$$Z = -0.8333 < Z_c = -1.645, \Rightarrow Z > Z_c$$

~~As $Z < Z_c$~~ $\therefore$ Do not reject H_0

Step 5: Compare probability (P-value) with α (significant level)

$$Z = -0.8333; \text{ left tailed probability } P(Z < -0.8333) = 0.2023$$

$$\left. \begin{array}{l} \text{P-value} = 0.2023 \\ \alpha = 0.05 \end{array} \right\} \text{P-value} > \alpha \Rightarrow \text{Do not Reject } H_0$$

Step 6: Conclusion: There is insufficient evidence at 5% level of significant, to conclude that the average box < 368g.

2-tailed Z-Test

Q) A machine fills a cereal boxes; the label claim $\mu = 368g$, population std. dev. $\sigma = 15g$. The quality inspector take sample $n = 25$ boxes and found $\bar{x} = 372.5g$.

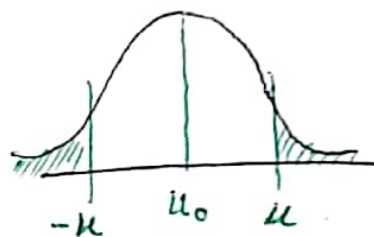
© Test whether the machine is filling acceptable at $\alpha = 0.05$, assuming the cereal box weight is normally distributed.

Sol: $\mu = 368g$; $\sigma = 15g$; $n = 25$; $\bar{x} = 372.5g$; $\alpha = 0.05$

Step 1: Purpose Hypothesis

$$H_0: \mu = 368g \quad (\mu = \mu_0)$$

$$H_A: \mu \neq 368g \quad (\mu \neq \mu_0)$$



Step 2: Decide the test

As $\mu \neq \mu_0 \Rightarrow$ Two tailed test

Step 3: Calculate Z & Z_c

$$(i) Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{372.5 - 368}{15 / \sqrt{25}}$$

$$= \frac{4.5}{15/3} = \frac{4.5}{3} = 1.5; \quad \boxed{Z = 1.5}$$

Step 4: Make a decision

$$Z = 1.5; \quad Z_c (-1.96 \text{ \& } +1.96)$$

$$\text{As } Z_{-c} < Z < Z_{+c}$$

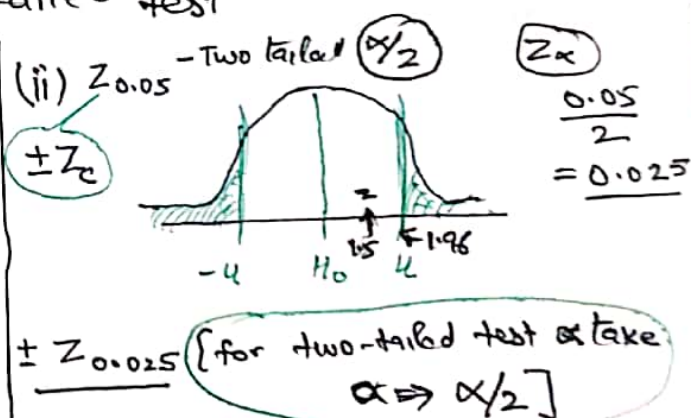
$\Rightarrow$ Z-value doesn't fall in the rejection region

$\therefore$ Fail to Reject H_0

Step 5: Compare p-value $P(Z > 1.5) \times 2$ ($\times 2$ because it is 2-tailed)

$$\Rightarrow \boxed{P = 0.1336} \quad \boxed{\alpha = 0.05} \quad = (1 - 0.9332) \times 2 = (0.0668) \times 2 = 0.1336$$

As $\boxed{p\text{-value} > \alpha} \Rightarrow \boxed{\text{Fail to reject } H_0}$ {Step 6: conclusion}



$$\text{From table } Z_{0.025} = 1.96$$

So critical values are

$$\underline{-1.96 \text{ and } +1.96}$$

End of Unit 2

CSE552: Business Analytics

Inferential Statistics and Hypothesis Testing

"In God we trust, all others bring data." — W. Edwards Deming